SCALE: Sharpness-Aware Correction for Addressing Long-Tail Effects in Combinatorial Medication Recommendation

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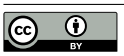
Abstract

With the widespread adoption of electronic health records, Combination Medication Recommendation (CMR) has become a key task in personalized medicine, aiming to generate safe and effective medication combinations based on patients' conditions and histories. However, existing methods often overlook the long-tailed distribution common in clinical data, where a few frequent diseases (head classes) dominate, while numerous rare ones (tail classes) have limited samples. This imbalance hinders the generalization of CMR models to tail diseases, limiting clinical applicability. We present the first systematic study of the long-tail problem in CMR. We observe that under imbalanced data, stochastic gradient descent tends to stagnate at negative curvature saddle points associated with tail diseases, degrading generalization. Moreover, the commonly used drug–drug interaction loss further worsens performance on tail classes due to its second-order structure. To tackle these issues, we propose Sharpness-Aware Correction for Addressing Long-Tail Effects (SCALE), a framework that integrates fine-grained-weighted sharpness-aware minimization to balance optimization between head and tail diseases. We theoretically analyze how SCALE mitigates the long-tail problem and derive generalization error bounds to validate its soundness. Extensive experiments show that SCALE significantly enhances generalization on rare diseases while maintaining strong recommendation performance on common ones.

CCS Concepts

• **Information systems** → **Recommender systems**; • **Applied computing** → **Health informatics**.

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Keywords

Recommender System, Combinatorial Medication Recommendation, Electronic Health Records.

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1 Introduction

With the growing volume of medical and health data, medication recommendation systems play an increasingly vital role in personalized medicine. For example, in chronic disease management, analyzing patients' Electronic Health Records (EHRs) and medical history enables these systems to recommend appropriate medication types and dosages for diabetic patients, aiding clinical decisions and enhancing treatment outcomes [1, 11, 34, 37, 47]. However, traditional methods mainly focus on individual medication recommendations and often fail to address the combination needs of patients with multimorbidity or complex conditions [28, 31]. They typically neglect Drug–Drug Interactions (DDI) and struggle to ensure the efficacy and safety of recommended combinations in clinical practice.

To address this challenge, Combinatorial Medication Recommendation (CMR) has emerged [30, 39]. By jointly considering patients' health status and medical history, CMR aims to provide accurate and safe prescriptions, maximizing therapeutic efficacy while minimizing adverse effects such as DDIs [10, 38, 40, 50]. With advanced data processing capabilities, CMR has become a key clinical tool, widely used in practice with real-world EHR data to support personalized medication combinations and complex treatment regimens [30, 39]. Additionally, advances in deep learning and Graph Neural Networks (GNNs), models such as GAMENet [35], have further improved the performance of medication recommendation systems, advancing the integration of CMR into clinical decision support.

Although substantial progress has been achieved in CMR research, the long-tail problem remains largely underexplored. In

real-world medication recommendation scenarios, disease prevalence varies considerably, leading to long-tail distributions in CMR datasets, where a few common diseases (head classes) dominate the samples, and numerous rare diseases (tail classes) are underrepresented. Models trained on such imbalanced data often exhibit poor generalization to tail classes, thereby degrading recommendation performance. As illustrated in Fig. 1, we analyze disease distributions in the MIMIC-III [19] and MIMIC-IV [18] datasets and evaluate several recent CMR baselines. The results indicate that: i) more than 80% of the diseases account for less than 20% of the total occurrences, indicating a distinct long-tail distribution pattern; and ii) model performance on rare diseases (e.g., auditory hallucinations) is substantially inferior to that on common diseases (e.g., hypertension), with an average gap of 22.9%. This performance imbalance severely limits the applicability of CMR models.

To address this challenge, we propose **S**harpness-Aware **C**orrection for **A**ddressing **L**ong-Tail **E**ffects (**SCALE**), a framework designed to enhance rare disease medication recommendation while preserving performance on common diseases. We jointly model the CMR prediction loss and the DDI safety loss by treating disease groups as long-tail classes and defining class-conditional risks. Based on the negative-curvature saddle point theory of the class-conditional Hessian spectrum, we reveal that *rare disease groups are more likely to stagnate at stable saddle points with negative curvature*, resulting in inferior generalization. Furthermore, *the composite second-order structure of the DDI loss introduces additional negative curvature*, making tail groups more prone to highly nonlinear regions and further exacerbating generalization degradation. To mitigate these issues, we adopt a Sharpness-Aware Minimization (SAM) strategy that minimizes class-conditional loss under worst-case neighborhood perturbations, facilitating escape from negative curvature directions and convergence to flatter minima. We theoretically show that SAM amplifies the second-order moment of gradient projections along negative curvature directions, thereby accelerating saddle point escape. To specifically benefit tail classes without overflattening head classes, we incorporate fine-grained weighting into the sharpness term, achieving balanced optimization. Finally, we introduce a sharpness increment mechanism to reduce backpropagation overhead, ensuring scalability and efficiency on large-scale datasets and models.

Our main contributions are summarized as follows:

- To our knowledge, this is the first work to systematically investigate the long-tail problem in CMR, revealing that it stems not only from disease group imbalance in the input data but also from the structural properties of the DDI loss.
- We propose SCALE, a unified framework that simultaneously addresses both sources of long-tail effects in CMR while preserving recommendation performance on common diseases.
- We provide theoretical insights into the mechanism by which SCALE alleviates the long-tail problem, showing that it amplifies the second moment of gradient projections along negative curvature directions and deriving a generalization error bound.
- Extensive experiments on multiple real-world datasets and diverse CMR models demonstrate that SCALE substantially improves generalization on tail diseases without sacrificing performance on head diseases.

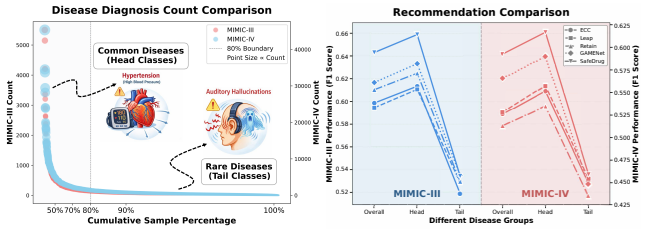


Figure 1: The left chart compares the frequencies of different diseases, while the right chart evaluates the recommendation of the current CMR model across disease groups, using the F1 score to balance precision and recall. For details on disease grouping, refer to the experimental section.

2 Related Work

2.1 Medication Recommendation

Combinatorial Medication Recommendation (CMR) aims to generate precise and safe prescriptions by jointly modeling patients' health conditions [29, 38, 40, 50]. Existing studies can be broadly categorized into instance-based and longitudinal methods. Instance-based methods focus on patients' current health status [9, 16, 33]. For example, LEAP formulates the task as a multi-instance multi-label decision process by modeling label–instance mappings within the current visit [48]. However, such methods typically ignore historical patient information, limiting their ability to ensure recommendation safety and effectiveness [15, 27, 43, 51]. In contrast, longitudinal methods incorporate temporal information from patients' hospitalization histories to capture disease progression and have gradually become dominant [21, 29, 41, 42, 45]. For instance, RETAIN employs a reverse-time attention mechanism to identify influential past visits and clinical features [7], while GAMENet leverages memory networks and DDI relationships to model historical medication patterns, improving recommendation performance [35].

Despite these advances, existing CMR methods overlook the long-tail nature of EHR data, resulting in poor generalization for rare diseases and limited recommendation effectiveness.

2.2 Long-tail Learning

Long-tail learning focuses on training models under highly imbalanced class distributions, aiming to improve generalization for minority classes while balancing performance between head and tail classes [44, 49]. Existing approaches are mainly categorized into resampling and reweighting methods. Resampling mitigates data imbalance by adjusting class frequencies, including oversampling minority classes and undersampling majority classes [5, 6, 36]. However, oversampling may cause overfitting on tail classes, whereas undersampling discards valuable data and degrades overall generalization. Reweighting methods instead assign class-dependent weights in the loss function to emphasize tail samples [2, 17, 26].

However, existing long-tail learning methods are not directly applicable to CMR, as they typically address label-level imbalance, whereas CMR involves input-level long-tail distributions over disease groups. The former aligns with a weighted empirical risk minimization problem, while the latter reflects conditional distribution

imbalance and limited generalization. This fundamental difference limits the effectiveness of applying prior methods to CMR.

3 Preliminary

3.1 Architecture of CMR

Although numerous CMR frameworks have emerged, they generally consist of the following three modules [29, 35, 39, 45]:

- **Patient representation extraction.** This module encodes the diagnoses and procedures of a patient's t-th visit into a unified representation $\boldsymbol{Xt} = f_x(\{X_i^d, X_i^p\}_{i=1}^t)$, where f_x is usually implemented with multiple Multi-Layer Perceptions (MLPs). Here, $X_i^d \in \{0,1\}^{n_d}$ and $X_i^p \in \{0,1\}^{n_p}$ denote the one-hot diagnosis and procedure vectors, respectively.
- **Medication representation extraction.** This module maps multimodal medication information, including textual descriptions M_T, identifiers M_I, and molecular structures M_S, into a unified medication representation $\boldsymbol{M} = f_m(M_T, M_I, M_S)$. The encoder f_m typically consists of modality-specific components (e.g., Pre-trained Language Models (PLMs), ID encoders, and GNNs), followed by representation alignment and fusion.
- **Combinatorial medication recommendation.** Using the historical information $\boldsymbol{Z} = f_c(\boldsymbol{X}_{t-1}, \boldsymbol{M}_{t-1})$ and current patient representation $\boldsymbol{X}_t$, where f_c denotes the combination function, this module predicts the medication combination for the current visit as $\boldsymbol{M}_t = f_{\text{cmr}}(\boldsymbol{X}_t, \boldsymbol{Z})$, where f_{cmr} implemented using MLPs.

While the specific architectures vary across studies, the functional roles of these modules remain largely consistent.

To train the above modules, CMR studies typically employ the following overall objective:

$$\mathcal{L}_{\text{Total}}(\boldsymbol{\theta}; \boldsymbol{M}_t, \boldsymbol{O}_t) = \mathcal{L}_{\text{Pred}}(\boldsymbol{\theta}; \boldsymbol{M}_t, \boldsymbol{O}_t) + \alpha \mathcal{L}_{\text{DDI}}(\boldsymbol{\theta}; \boldsymbol{M}_t), \quad (1)$$

where $\boldsymbol{\theta}$ denotes the model parameters and $\boldsymbol{O}_t$ is the ground-truth medication combination. The prediction loss $\mathcal{L}_{\text{Pred}}$ is formulated as a multi-label classification loss, commonly implemented using Binary Cross-Entropy (BCE):

$$\mathcal{L}_{\text{Pred}}(\boldsymbol{\theta}; \boldsymbol{M}_t, \boldsymbol{O}_t) = -\sum_{i=1}^{|\mathcal{M}|} \left(O_t^i \log M_t^i + (1 - O_t^i) \log(1 - M_t^i) \right), \quad (2)$$

where $\mathcal{M}$ denotes the medication set, and $O_t^i \in \{0,1\}$ and $M_t^i \in \{0,1\}$ represent the ground-truth label and predicted probability for the i-th medication, respectively. The DDI loss $\mathcal{L}_{\text{DDI}}$ quantifies the interaction risk within a medication combination and is typically defined as:

$$\mathcal{L}_{\text{DDI}}(\boldsymbol{\theta}; \boldsymbol{M}_t) = \sum_{i=1}^{|\mathcal{M}|} \sum_{j=1}^{|\mathcal{M}|} \left(M_t^i (M_t^j)^\top \right) \cdot \boldsymbol{A}_{ij}, \quad (3)$$

where $\boldsymbol{A}_{ij}$ encodes the interaction between medications i and j, derived from real-world DDI knowledge. The coefficient α balances recommendation accuracy and medication safety by trading off between the prediction and DDI losses.

3.2 Definition of Long-Tail Problem in CMR

To formalize the long-tail problem in CMR, we first define disease groups. Let $g \in \mathcal{G}$ denote a disease group, which can be defined as a single disease or a set of diseases. For clarity, we focus on the single-disease case at the t-th time point in the following analysis. The training set is denoted as $\mathcal{S} = \{\{(\boldsymbol{X}_t^{(i)}, \boldsymbol{M}_{t-1}^{(i)}, \boldsymbol{O}_t^{(i)}, g^{(i)})\}_{t=1}^{T_i}\}_{i=1}^n$, where T_i is the number of visits for patient i, n is the total number of samples, and $\mathcal{S} \sim \mathcal{D}$ follows the training distribution. Let n_g denote the number of samples in group g, and $\pi_g = n_g/n$ its proportion in the training set. Without loss of generality, the disease groups can be partitioned into head and tail subsets, $\mathcal{G}_{\text{head}}$ and $\mathcal{G}_{\text{tail}}$, such that $\pi_{g_h} \gg \pi_{g_t}$ for all $g_h \in \mathcal{G}_{\text{head}}$ and $g_t \in \mathcal{G}_{\text{tail}}$.

The empirical loss over the training set $\mathcal{S}$ is defined as:

$$\hat{\mathcal{L}}_{\mathcal{S}}(\boldsymbol{\theta}) = \frac{1}{n} \sum_{i=1}^{n} \left(\frac{1}{T_i} \sum_{t=1}^{T_i} \mathcal{L}_{\text{Total}} \left(\boldsymbol{\theta}; f_{\text{cmr}}(\boldsymbol{X}_t^{(i)}, \boldsymbol{M}_{t-1}^{(i)}), \boldsymbol{O}_t^{(i)} \right) \right), \quad (4)$$

which corresponds to optimization under the imbalanced training distribution. The group-conditional empirical loss for disease group g is given by:

$$\hat{\mathcal{L}}_{\mathcal{S}}^{g}(\boldsymbol{\theta}) = \frac{1}{n} \sum_{g^{(i)}=g} \left(\frac{1}{T_i} \sum_{t=1}^{T_i} \mathcal{L}_{\text{Total}} \left(\boldsymbol{\theta}; f_{\text{cmr}}(\boldsymbol{X}_t^{(i)}, \boldsymbol{M}_{t-1}^{(i)}), \boldsymbol{O}_t^{(i)} \right) \right). \quad (5)$$

We further denote the expected losses under the true distribution $\mathcal{D}$ and the balanced distribution $\mathcal{D}_{\text{bal}}$ as $\mathcal{L}_{\mathcal{D}}(\boldsymbol{\theta})$ and $\mathcal{L}_{\mathcal{D}_{\text{bal}}}(\boldsymbol{\theta})$, respectively. **Our objective is to learn parameters $\boldsymbol{\theta}$ from the imbalanced dataset $\mathcal{S}$ that minimize $\mathcal{L}_{\mathcal{D}_{\text{bal}}}(\boldsymbol{\theta})$, thereby achieving strong performance on a balanced test distribution.**

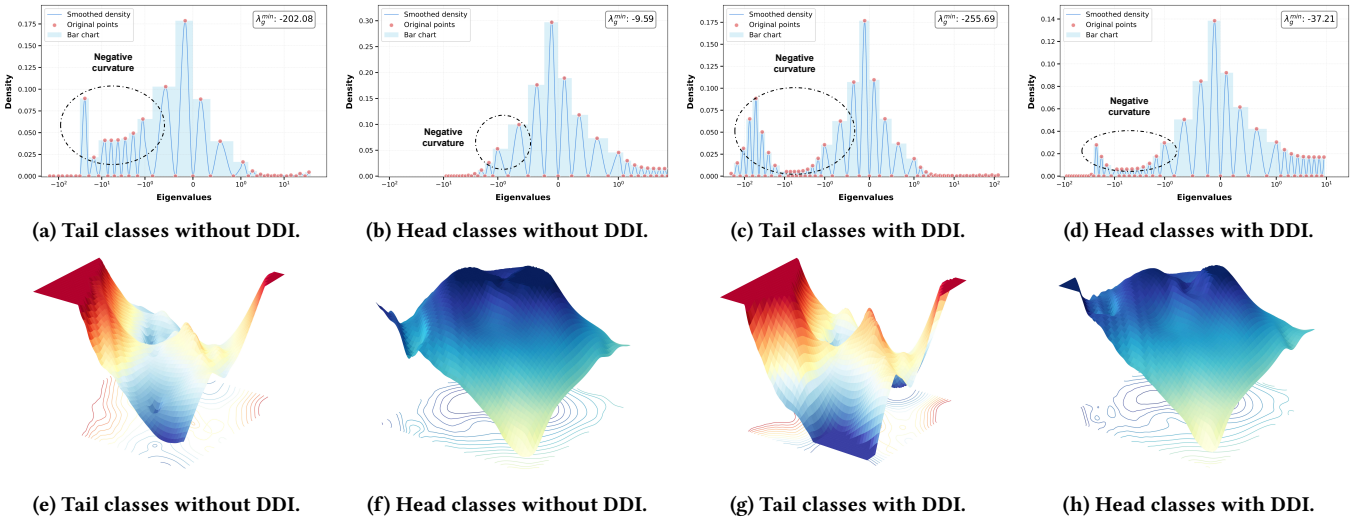
4 Methodology

In this section, we analyze the root causes of the long-tail problem in CMR from the perspective of negative-curvature saddle point theory. Based on this analysis, we propose SCALE, a sharpness-aware optimization framework that promotes escape from saddle points along negative curvature directions and convergence to flatter solutions. This design enables a balanced optimization between tail and head groups. We further provide a comprehensive theoretical analysis along with a practical optimization implementation.

4.1 Causes of the Long-Tail Problem in CMR

Prior studies on optimization theory have shown that solutions located at negative-curvature saddle points or sharp regions of the loss landscape often exhibit poor generalization [8, 13, 20, 23, 32]. Motivated by this insight, we analyze the average loss behavior of CMR models trained on class-imbalanced data across different disease groups. Specifically, we characterize the loss landscapes of head and tail classes by computing the Hessian eigenvalue (spectral) density. As the Hessian spectrum reflects the distribution of positive and negative curvatures, its central tendency reveals key geometric properties of the underlying loss surface. We further report class-wise validation accuracy for different disease groups g to complement the spectral analysis.

We estimate the Hessian spectral density using the Lanczos algorithm [14] and conduct experiments on the MIMIC-IV dataset with recent CMR models SafeDrug. As illustrated in Fig. 2, class-wise Hessian analysis shows that models trained on imbalanced data tend to converge to solutions with pronounced negative curvature for tail class losses, indicating a higher likelihood of saddle point convergence. In contrast, head classes are more likely to converge to local minima. Moreover, the loss landscapes associated with tail



(a) Tail classes without DDI. (b) Head classes without DDI. (c) Tail classes with DDI. (d) Head classes with DDI.

(e) Tail classes without DDI. (f) Head classes without DDI. (g) Tail classes with DDI. (h) Head classes with DDI.

Figure 2: Through Hessian analysis of the head and tail class losses, we observe that training the CMR network on the MIMIC-III and MIMIC-IV datasets leads to the tail class converging to a region of large negative curvature, indicating a saddle point. In contrast, the head class converges to a flatter minimum. Furthermore, enforcing the DDI loss during training reduces drug conflicts in recommendations but increases the likelihood of the model entering the saddle point region.

classes are significantly sharper than those of head classes. These observations suggest that, under data imbalance, CMR models are more prone to reside in negative-curvature or sharp regions of the loss landscape for rare diseases, leading to inferior generalization performance.

We further perform ablation studies to assess the impact of removing the DDI loss on the Hessian spectral density. The results show that incorporating the DDI loss substantially increases both the magnitude and density of negative eigenvalues for tail classes, yielding a markedly sharper loss landscape. Analysis of the DDI loss reveals that, due to the interaction between sigmoid saturation and network nonlinearity, its composite second-order structure introduces a systematic negative curvature component. As tail classes are more likely to activate this mechanism, the DDI loss not only fails to mitigate the long-tail problem but instead amplifies the generalization risk for rare diseases from an optimization geometry perspective. The corresponding theoretical derivation is provided in Appendix A. Overall, these findings indicate that the DDI loss drives tail groups toward highly nonlinear negative-curvature regions, further widening the generalization gap between head and tail classes.

4.2 Fine-grained Weighted SAM

The above analysis indicates that mitigating the long-tail problem in CMR, which arises from both disease group imbalance as well as the amplification effect introduced by the DDI loss, ultimately requires guiding the optimization process so that stochastic gradients preserve non-zero components in directions associated with negative curvature. This enables the training process to escape negative-curvature saddle points and converge toward flatter regions of the loss landscape, thereby improving generalization.

4.2.1 Introducing SAM into CMR. Motivated by the above insights, we adopt Sharpness-Aware Minimization (SAM), which explicitly captures and regulates the geometry of the loss landscape [25, 52, 53], thereby improving the model's generalization ability. By minimizing the worst-case loss within a local neighborhood, SAM amplifies gradient components along directions of negative curvature, thereby facilitating escape from saddle points. Accordingly, we apply SAM to each class-conditional loss $\hat{\mathcal{L}}_{\mathcal{S}}^{g}(\boldsymbol{\theta})$ and formulate the SAM-based objective as follows:

$$\hat{\mathcal{L}}_{\mathcal{S}}^{\mathrm{SAM}}(\boldsymbol{\theta}) = \sum_{g \in \mathcal{G}} \max_{\|\boldsymbol{\varepsilon}\|^2 \le \rho} \hat{\mathcal{L}}_{\mathcal{S}}^{g}(\boldsymbol{\theta} + \boldsymbol{\varepsilon}), \tag{6}$$

where ρ denotes the perturbation radius. To analyze the optimization behavior at the group level, we examine the gradient discrepancy between the original and SAM objectives under stochastic gradient descent, yielding the following result.

Theorem 4.1 (**Avoiding Saddle Points**). *Define the SAM gradient for group g as $\nabla_{\boldsymbol{\theta}} \hat{\mathcal{L}}_{\mathcal{S}}^{g,\mathrm{SAM}}(\boldsymbol{\theta}) = \nabla_{\boldsymbol{\theta}} \hat{\mathcal{L}}_{\mathcal{S}}^{g}(\boldsymbol{\theta} + \boldsymbol{\varepsilon}_g^{\star})$, where $\boldsymbol{\varepsilon}_g^{\star} = \arg\max_{\|\boldsymbol{\varepsilon}\|^2 \le \rho} \hat{\mathcal{L}}_{\mathcal{S}}^{g}(\boldsymbol{\theta} + \boldsymbol{\varepsilon})$. Let $\boldsymbol{H}_{\mathcal{S}}^{g}(\boldsymbol{\theta}) = \nabla_{\boldsymbol{\theta}}^2 \hat{\mathcal{L}}_{\mathcal{S}}^{g}(\boldsymbol{\theta})$ be the Hessian of the class-conditional loss, with smallest eigenvalue $\lambda_g^{\min}$ and corresponding unit eigenvector $\boldsymbol{v}_g(\boldsymbol{\theta})$. Then, we have*

$$\mathbb{E}\left[\langle \boldsymbol{v}_g, \nabla_{\boldsymbol{\theta}} \hat{\mathcal{L}}_{\mathcal{S}}^{g,\mathrm{SAM}}(\boldsymbol{\theta})\rangle^2\right] = (1 + \rho\lambda_g^{\min})^2 \mathbb{E}\left[\langle \boldsymbol{v}_g, \nabla_{\boldsymbol{\theta}} \hat{\mathcal{L}}_{\mathcal{S}}^{g}(\boldsymbol{\theta})\rangle^2\right]. \tag{7}$$

Theorem 4.1 shows that SAM amplifies the stochastic gradient component along the most negative curvature direction by a factor of $(1 + \rho\lambda_g^{\min})^2$, thereby accelerating escape from saddle points in the group-conditional loss landscape and improving generalization. Additionally, it can be observed that a larger coefficient ρ further strengthens this effect. The full proof is provided in Appendix B.

4.2.2 Introducing fine-grained weighting mechanism. However, SAM alone is insufficient to fully resolve the long-tail problem, as its flattening effect is still primarily influenced by the number of samples per class, potentially placing insufficient emphasis on tail classes while excessively flattening head classes. To address this limitation, we introduce a fine-grained weighting mechanism that modulates the strength of SAM across disease groups:

$$\hat{\mathcal{L}}_{\mathcal{S}}^{\mathrm{W-SAM}}(\boldsymbol{\theta}) = \sum_{g\in\mathcal{G}} \psi_g \cdot \max_{\|\boldsymbol{\varepsilon}\|^2\le\rho} \hat{\mathcal{L}}_{\mathcal{S}}^{g}(\boldsymbol{\theta}+\boldsymbol{\varepsilon}), \tag{8}$$

where ψ_g controls the sharpness intensity for group g, assigning larger weights to tail classes and smaller (but non-zero) weights to head classes, thereby balancing optimization between them.

Specifically, ψ_g is designed to increase as the class frequency π_g decreases. We consider the following weighting schemes:

- **Inverse Frequency Weight (IFW):** $\psi_g = (\pi_g + \epsilon)^{-1}$, where ϵ ensures numerical stability;
- **Logarithmic Weight (LogW):** $\psi_g = \log\left(1 + \pi_g^{-1}\right)$, which prevents excessively large weights;
- **Exponentially Decaying Weight (ExpW):** $\psi_g = (1-\pi_g)^{\beta}$, where $\beta > 0$ controls the decay rate.

4.2.3 Improving optimization efficiency. Directly optimizing Eq. (8) is computationally prohibitive, as each update requires evaluating class-wise gradients at perturbed parameters $(\boldsymbol{\theta}+\boldsymbol{\varepsilon}_g^{\star})$, incurring at least $|\mathcal{G}|$ backward passes. To improve efficiency, we introduce a sharpness increment mechanism, $\Delta_g(\boldsymbol{\theta},\boldsymbol{\varepsilon}) \triangleq \hat{\mathcal{L}}_{\mathcal{S}}^{g}(\boldsymbol{\theta}+\boldsymbol{\varepsilon}) - \hat{\mathcal{L}}_{\mathcal{S}}^{g}(\boldsymbol{\theta})$, and optimize the aggregated increment instead of the SAM objective itself. This reformulates Eq. (8) as:

$$\mathcal{T}_{\mathcal{S}}(\boldsymbol{\theta}) = \hat{\mathcal{L}}_{\mathcal{S}}(\boldsymbol{\theta}) + \gamma \cdot \max_{\|\boldsymbol{\varepsilon}\|^2\le\rho} \sum_{g\in\mathcal{G}} \psi_g \cdot \Delta_g(\boldsymbol{\theta},\boldsymbol{\varepsilon}), \tag{9}$$

where γ controls the contribution of sharpness. Let $\hat{\mathcal{L}}_{\mathcal{S}}^{\gamma}(\boldsymbol{\theta}) \triangleq \sum_{g\in\mathcal{G}} \psi_g \hat{\mathcal{L}}_{\mathcal{S}}^{g}(\boldsymbol{\theta})$ and the inner problem is $\hat{\mathcal{L}}_{\mathcal{S}}^{\mathrm{In}} \triangleq \max_{\|\boldsymbol{\varepsilon}\|^2\le\rho} \sum_{g\in\mathcal{G}} \psi_g \cdot \Delta_g(\boldsymbol{\theta},\boldsymbol{\varepsilon})$. To solve Eq. (9), we apply a first-order Taylor approximation to the inner maximization, yielding:

$$\boldsymbol{\varepsilon}^{\star}(\boldsymbol{\theta}) \approx \arg\max_{\|\boldsymbol{\varepsilon}\|_2\le\rho} \boldsymbol{\varepsilon}^{\top}\nabla_{\boldsymbol{\theta}}\hat{\mathcal{L}}_{\mathcal{S}}^{\gamma}(\boldsymbol{\theta}) = \rho\frac{\nabla_{\boldsymbol{\theta}}\hat{\mathcal{L}}_{\mathcal{S}}^{\gamma}(\boldsymbol{\theta})}{\|\nabla_{\boldsymbol{\theta}}\hat{\mathcal{L}}_{\mathcal{S}}^{\gamma}(\boldsymbol{\theta})\|_2}. \tag{10}$$

Substituting this approximation into the outer objective leads to:

$$\nabla_{\boldsymbol{\theta}}\mathcal{T}(\boldsymbol{\theta}) \approx \nabla_{\boldsymbol{\theta}}\left(\left[\hat{\mathcal{L}}_{\mathcal{S}}(\boldsymbol{\theta}) - \gamma\hat{\mathcal{L}}_{\mathcal{S}}^{\gamma}(\boldsymbol{\theta})\right] + \gamma\hat{\mathcal{L}}_{\mathcal{S}}^{\gamma}(\boldsymbol{\theta}+\boldsymbol{\varepsilon}^{\star}(\boldsymbol{\theta}))\right). \tag{11}$$

As summarized in Algorithm 1, each parameter update requires only three backward passes: computing $\nabla_{\boldsymbol{\theta}}\hat{\mathcal{L}}_{\mathcal{S}}^{\gamma}(\boldsymbol{\theta})$, $\nabla_{\boldsymbol{\theta}}[\hat{\mathcal{L}}_{\mathcal{S}}(\boldsymbol{\theta}) - \gamma\hat{\mathcal{L}}_{\mathcal{S}}^{\gamma}(\boldsymbol{\theta})]$, and $\nabla_{\boldsymbol{\theta}}\hat{\mathcal{L}}_{\mathcal{S}}^{\gamma}(\boldsymbol{\theta}+\boldsymbol{\varepsilon}^{\star}(\boldsymbol{\theta}))$, where the first two steps essentially require only a single gradient computation. Therefore, SCALE achieves high computational efficiency and is well-suited for large-scale datasets and models with substantial parameter counts.

4.3 Theoretical Analysis

We further analyze the effectiveness of SCALE from a theoretical perspective. We assume that $\mathcal{L}_{\mathrm{Total}}(\boldsymbol{\theta}; f_{\mathrm{cmr}}(\boldsymbol{X}_t^{(i)}, \boldsymbol{M}_{t-1}^{(i)}), \boldsymbol{O}_t^{(i)})$ is upper bounded by a constant ξ, a standard and practical assumption. Under this setting, we derive the following generalization error bound. The detailed proof is provided in Appendix C.

Algorithm 1 SCALE Algorithm

Input: Training set $\mathcal{S}$, perturbation radius ρ, smoothing coefficient β, sharpness weight γ, learning rate η, the maximum number of iterations $E_{\max}$.
Output: Trained model parameters $\boldsymbol{\theta}$.

```
Initialize model parameters θ_0, e = 0
while not converged do
    Sample mini-batch B = {{(X_t^(i), M_{t-1}^(i), O_t^(i), g^(i))}_{t=1}^{T_i}}_{i=1}^{b}.
    NOTE: B is a visit in certain CMR models.
    Compute the CMR network training loss L̂_B(θ_e).
    Compute the sharpness incremental loss L̂_B^γ(θ_e).
    Compute the gradient ∇_θ L̂_B^γ(θ_e) and compute
    perturbation ε*(θ_e) based on Eq.(10).
    Compute gradient g_1 = ∇_θ[L̂_B(θ) − γ L̂_B^γ(θ))] |_{θ_e}.
    Compute gradient g_2 = ∇_θ L̂_B^γ(θ) |_{θ_e+ε*(θ_e)}.
    Update parameters: θ_{e+1} = θ_e − η(g_1 + γ g_2).
    Update iteration count: e ← e + 1.
end while
```

return Optimized model parameters $\boldsymbol{\theta}_{E_{\max}}$.

Theorem 4.2 (**Generalization bound of SCALE**). *Let $\pi_{\min} = \min_{g\in\mathcal{G}}\pi_g$, $\hat{\xi} = \sum_{g\in\mathcal{G}}\psi_g\pi_g\xi$, and let $\boldsymbol{H}_{\mathcal{D}}(\boldsymbol{\theta})$ denote the Hessian of $\mathcal{L}_{\mathcal{D}}(\boldsymbol{\theta})$. Assume the balanced test distribution $\mathcal{D}_{\mathrm{bal}}$ assigns equal probability to each disease group. Then, for any $\gamma \ge 0$ and $\rho > 0$, with probability at least $1-\delta$ over $\mathcal{S}\sim\mathcal{D}$,*

$$\begin{aligned}\mathcal{L}_{\mathcal{D}_{\mathrm{bal}}}(\boldsymbol{\theta}) \;\le\; & \frac{2}{|\mathcal{G}|\pi_{\min}}\mathcal{T}_{\mathcal{S}}(\boldsymbol{\theta}) - \frac{\gamma\rho^2}{2|\mathcal{G}|\pi_{\min}\left(\sqrt{d}+\sqrt{2\ln n}\right)^2}\mathrm{Tr}\big(\boldsymbol{H}_{\mathcal{D}}(\boldsymbol{\theta})\big) \\ & + \frac{1}{|\mathcal{G}|\pi_{\min}}\left[\frac{40(\xi+\gamma\hat{\xi})}{3n}\ln\frac{2}{\delta} + \frac{\gamma C(\boldsymbol{\theta},\rho,d,n,\delta)}{n}\right] \\ & - \frac{\gamma}{|\mathcal{G}|\pi_{\min}}\cdot\tilde{O}\Big(\tfrac{d\rho^2}{(\sqrt{d}+\sqrt{2\ln n})^2}\Big),\end{aligned} \tag{12}$$

where d denotes the number of model parameters.

From Theorem 4.2, we draw the following insights:

- **Mechanism of SCALE.** In highly imbalanced CMR settings, the dominant term in the balanced test risk scales with $\pi_{\min}$, implying that minimizing empirical risk or applying first-order reweighting alone cannot theoretically resolve the long-tail problem. SCALE explicitly suppresses the Hessian trace through a fine-grained weighted sharpness-aware objective, reducing negative curvature for tail disease solutions. This curvature suppression counteracts the amplification effect induced by small $\pi_{\min}$, providing a theoretical explanation for SCALE's strong generalization on rare diseases without degrading common diseases' recommendation performance.
- **Effect of hyperparameters.** The bound further indicates that γ and ρ play complementary roles. Increasing γ strengthens Hessian trace suppression for tail groups, while an appropriate ρ enhances curvature-aware regularization in high-dimensional spaces. However, overly large γ or ρ increases complexity-dependent terms, suggesting that moderate values yield the optimal trade-off between tail generalization and overall optimization stability.

Table 1: Statistics of the datasets used in our experiments.

Dataset	Patients	Visits	Diagnoses	Procedures	Medications
MIMIC-III	6350	15032	1958	1430	112
MIMIC-IV	77387	208148	2000	13324	127

5 Experiment

To evaluate the proposed framework SCALE, we conduct extensive experiments on real-world medical datasets. The experiments are designed to answer the following four Research Questions (RQs):

- **RQ1:** Can SCALE enhance medication recommendation for rare diseases without sacrificing performance on common diseases?
- **RQ2:** How computationally efficient is SCALE, and can it scale to large datasets and high-capacity models?
- **RQ3:** How do different hyperparameter settings influence the performance of SCALE?
- **RQ4:** What mechanisms underlie SCALE, and which components of the model are fundamentally affected?

5.1 Experimental Settings

5.1.1 Datasets. We evaluate the proposed and baseline methods on two real-world EHR datasets, MIMIC-III [19] and MIMIC-IV [18]. Both datasets are widely adopted in prior studies [7, 33, 35, 43, 48] and have been rigorously anonymized and preprocessed. Based on [43, 50], we select patients who had visited these two datasets at least twice and used the ATC level 3 code as the target label. Table 1 reports their statistics. Following the Pareto principle [4], we designate the top 20% of the most frequent diseases as head classes, while the remaining are tail classes. Since each sample in the CMR input contains multiple ICD codes, we count the number of head class diseases. If it exceeds 2/3, the sample is classified as a head sample; if below 1/3, it is classified as a tail sample. The weighted average of ψ_g for each disease is used as the sample's weight. Consistent with [22, 24, 45, 50], the dataset is split into training, validation, and test sets with a ratio of 2:1:1 for the recommendation task.

To evaluate the generality and adaptability of SCALE, we adopt five representative CMR models as backbone networks. These models can be broadly categorized into two groups:

- *Instance-based models.* **ECC** [33] formulates CMR as a multi-label classification problem using an ensemble learning framework. **LEAP** [48] models the task as a multi-instance decision process by integrating label instances with current visit information.
- *Longitudinal models.* **RETAIN** [7] employs a reverse-time attention RNN to model longitudinal patient visits and predict future medications. **GAMENet** [35] leverages a memory-augmented neural network to capture historical prescription patterns. **SafeDrug** [43] improves recommendation accuracy and safety by incorporating drug molecular structure information.

5.1.2 Baseline methods. We use the original CMR models and the basic version of SAM as baselines. Conventional long-tail learning methods based on resampling or re-weighting are excluded, as their direct application violates key theoretical assumptions for this task. A detailed discussion is provided in Section 2.2.

5.1.3 Evaluation protocols. Following prior work [43, 45], we adopt standard metrics to evaluate consistency with real-world prescription patterns, including Jaccard similarity, average F1 score (F1), PRAUC, DDI rate (DDI), and the average number of medications per prescription (AvgMed). In addition, the average training time per epoch is reported to assess computational efficiency.

5.1.4 Implementation details. All models are trained with the Adam optimizer, and hyperparameters are tuned on the training set via grid search. The embedding dimension, and learning rate are set to 64 and 5e-4, respectively. We follow the recommended settings from the original papers and further fine-tune them for optimal performance. In our method, we set ρ, β, γ to 0.5, 10, 0.8, respectively. Experiments are conducted on a single machine with an NVIDIA GTX 4090 GPU. All results are averaged over five runs, with corresponding errors reported [1].

5.2 Results and Discussions

5.2.1 RQ1: Effectiveness of SCALE. We systematically evaluate the effectiveness of SCALE on five CMR models and two real-world EHR datasets. We compare overall recommendation performance before and after applying SCALE, as well as performance on head and tail disease classes. The results are summarized in Table 2, from which the following observations are drawn:

- **Overall performance.** Directly incorporating SAM into CMR models does not yield consistent improvements and often introduces notable side effects. This is because SAM performs group-level local optimization, where perturbation magnitude is still influenced by group size, leading to over-flattening on head classes and insufficient enhancement for tail classes. In contrast, SCALE achieves consistent and significant improvements across all datasets and backbone models. This advantage arises from (i) the fine-grained weighting mechanism, which enables precise control across different groups, and (ii) the sharpness aware optimization shift from training driven by local perturbations to training guided by global objectives, which substantially improves generalization.
- **Head vs. tail performance.** While SAM provides limited gains on tail classes, these improvements often come at the cost of degraded head-class performance. By contrast, SCALE yields notable improvements on tail classes, achieving average gains of 13.02% in MIMIC-III and 11.61% in MIMIC-IV, while largely preserving and in some cases slightly improving performance on head classes. This can be attributed to the avoidance of over-intervention on head classes and the global optimization nature, which induces beneficial cross-group interaction effects.

Overall, these results demonstrate that SCALE effectively alleviates long-tail performance degradation and achieves a more balanced trade-off between head and tail classes.

5.2.2 RQ2: Computational efficiency and scalability. We further evaluate the computational efficiency of different methods across CMR models and EHR datasets, using the average training time per epoch as the metric. As shown in Fig. 3, SCALE introduces only a modest overhead (1.52× on average) compared to the original models; for several backbones (e.g., GAMENet and ECC), the additional

[1] Our code is available at https://github.com/muyiahhh/SCALE.

Table 2: Performance comparison across datasets and CMR models. All values represent the averages of 5 experiments, with values in (·) showing the average change rate of the five metrics. The best performance is marked in bold. All results are analyzed using t-tests, with significant differences at $p < 0.05$.

Method	Overall Performance						Tail Performance						Head Performance					
	Jaccard (↑)	F1 (↑)	PRAUC (↑)	DDI (↓)	Med. (↑)	Average Change	Jaccard (↑)	F1 (↑)	PRAUC (↑)	DDI (↓)	Med. (↑)	Average Change	Jaccard (↑)	F1 (↑)	PRAUC) (↑)	DDI (↓)	Med. (↑)	Average Change
MIMIC-III																		
ECC	0.4418	0.5986	0.7063	0.0765	14.54	-	0.3558	0.5185	0.5704	**0.0705**	11.20	-	0.4519	0.6135	0.7259	0.0791	14.94	-
+SAM	0.4338	0.5884	0.6979	0.0750	14.34	(-0.81%)	0.3836	0.5543	0.6003	0.0753	11.90	(+3.87%)	0.4414	0.5993	0.7071	**0.0774**	14.56	(-1.51%)
+SCALE	**0.4490**	**0.6061**	**0.7240**	**0.0716**	**14.69**	**(+2.56%)**	**0.4143**	**0.5701**	**0.6672**	0.0736	**13.13**	**(+11.24%)**	**0.4581**	**0.6258**	**0.7410**	0.0809	**15.27**	**(+1.09%)**
LEAP	0.4312	0.5944	0.6337	0.0699	19.24	-	0.3453	0.5283	0.5552	0.0623	15.21	-	0.4484	0.6107	0.6643	**0.0725**	19.86	-
+SAM	0.4403	0.6068	0.6434	0.0711	19.61	(+1.18%)	0.3681	0.5592	0.5937	0.0656	16.22	(+4.14%)	0.4510	0.6158	0.6658	0.0728	19.95	(+0.33%)
+SCALE	**0.4469**	**0.6141**	**0.6529**	**0.0671**	**19.82**	**(+2.29%)**	**0.4024**	**0.6183**	**0.6290**	**0.0617**	**18.11**	**(+13.37%)**	**0.4566**	**0.6234**	**0.6711**	0.0740	**20.06**	**(+0.79%)**
RETAIN	0.4458	0.6104	0.7390	**0.0737**	18.47	-	0.3648	0.5292	0.5824	**0.0624**	14.91	-	0.4564	0.6234	0.7410	0.0809	19.27	-
+SAM	0.4537	0.6239	0.7554	0.0753	18.87	(+1.23%)	0.3890	0.5714	0.6237	0.0677	15.74	(+4.35%)	0.4583	0.6309	0.7562	0.0821	**19.85**	(+0.31%)
+SCALE	**0.4573**	**0.6396**	**0.7650**	0.0749	**19.45**	**(+2.91%)**	**0.4199**	**0.6197**	**0.6878**	0.0639	**17.82**	**(+14.06%)**	**0.4621**	**0.6405**	**0.7729**	**0.0750**	19.52	**(+1.80%)**
GAMENet	0.4563	0.6168	0.7135	0.0795	23.89	-	0.3731	0.5347	0.5739	**0.0845**	17.48	-	0.4763	0.6335	0.7448	0.0822	24.88	-
+SAM	0.4508	0.6060	0.7038	0.0785	23.47	(-0.97%)	0.4025	0.5687	0.6089	0.0910	18.46	(+3.65%)	0.4651	0.6179	0.7232	**0.0800**	24.25	(-1.53%)
+SCALE	**0.4743**	**0.6223**	**0.7536**	**0.0766**	**25.20**	**(+3.92%)**	**0.4448**	**0.6068**	**0.6695**	0.0881	**20.83**	**(+12.85%)**	**0.4812**	**0.6401**	**0.7606**	0.0818	**25.37**	**(+1.31%)**
SafeDrug	0.4835	0.6433	0.7401	0.0616	18.86	-	0.3732	0.5340	0.5826	0.0605	14.93	-	0.5057	0.6592	0.7646	0.0642	19.47	-
+SAM	0.4913	0.6516	0.7543	0.0628	19.23	(+0.98%)	0.3971	0.5675	0.6136	0.0648	16.09	(+3.74%)	0.5066	0.6652	0.7689	0.0648	19.64	(+0.34%)
+SCALE	**0.5121**	**0.6586**	**0.7633**	**0.0615**	**20.23**	**(+3.71%)**	**0.4466**	**0.6254**	**0.6883**	**0.0620**	**17.23**	**(+13.56%)**	**0.5164**	**0.6665**	**0.7824**	**0.0636**	**20.88**	**(+2.77%)**
MIMIC-IV																		
ECC	0.3901	0.5263	0.6687	0.0653	7.82	-	0.3396	0.4492	0.5679	0.0611	5.52	-	0.4087	0.5516	0.6993	0.0686	8.20	-
+SAM	0.3958	0.5194	0.6600	0.0646	7.70	(-0.29%)	0.3615	0.4799	0.6049	0.0630	5.92	(+4.79%)	0.4004	0.5364	0.6833	0.0697	8.02	(-2.16%)
+SCALE	**0.4042**	**0.5405**	**0.6717**	**0.0637**	**8.11**	**(+2.61%)**	**0.3775**	**0.5019**	**0.6313**	**0.0603**	**6.14**	**(+9.30%)**	**0.4156**	**0.5626**	**0.7131**	**0.0676**	**8.31**	**(+1.68%)**
LEAP	0.3791	0.5288	0.5480	0.0552	9.74	-	0.3285	0.4545	0.5071	0.0511	7.45	-	0.3966	0.5578	0.5717	**0.0580**	10.23	-
+SAM	0.3813	0.5163	0.5543	**0.0538**	9.59	(+0.06%)	0.3492	0.4837	0.5473	0.0501	8.01	(+6.06%)	0.3855	0.5425	0.5547	0.0593	9.96	(-2.67%)
+SCALE	**0.3908**	**0.5450**	**0.5745**	0.0576	**9.97**	**(+1.80%)**	**0.3630**	**0.5037**	**0.5593**	**0.0492**	**8.30**	**(+9.35%)**	**0.4016**	**0.5650**	**0.5828**	0.0587	**10.43**	**(+1.03%)**
RETAIN	0.3641	0.5125	0.6784	**0.0527**	9.93	-	0.3086	0.4349	0.5420	0.0618	8.71	-	0.3851	0.5347	0.7089	0.0818	10.46	-
+SAM	0.3567	0.5162	0.6860	0.0568	9.80	(-1.87%)	0.3245	0.4607	0.5794	0.0617	9.40	(+5.21%)	0.3751	0.5213	0.6917	**0.0796**	10.15	(-1.56%)
+SCALE	**0.3760**	**0.5334**	**0.7047**	0.0535	**10.13**	**(+2.36%)**	**0.3409**	**0.4811**	**0.5997**	**0.0602**	**9.65**	**(+9.02%)**	**0.3901**	**0.5337**	**0.7206**	0.0809	**10.73**	**(+1.09%)**
GAMENet	0.4138	0.5637	0.6679	0.0682	15.51	-	0.3348	0.4475	0.5541	0.0670	12.83	-	0.4348	0.5904	0.7078	0.0712	16.23	-
+SAM	0.4179	0.5719	0.6560	0.0674	15.13	(-0.21%)	0.3586	0.4744	0.5880	0.0661	13.53	(+5.23%)	0.4261	0.5741	0.6901	**0.0691**	15.81	(-1.29%)
+SCALE	**0.4321**	**0.5822**	**0.6891**	**0.0673**	**16.23**	**(+3.29%)**	**0.3966**	**0.5284**	**0.6591**	**0.0653**	**15.17**	**(+15.24%)**	**0.4419**	**0.6015**	**0.7059**	0.0699	**16.89**	**(+1.82%)**
SafeDrug	0.4418	0.5926	0.6788	0.0535	7.38	-	0.3511	0.4592	0.5025	0.0587	7.75	-	0.4662	0.6172	0.7092	0.0560	10.79	-
+SAM	0.4442	0.5807	0.6694	0.0529	10.27	(-0.55%)	0.3719	0.4915	0.5978	0.0590	8.26	(+4.87%)	0.4544	0.6032	0.6940	0.0545	10.73	(-1.31%)
+SCALE	**0.4632**	**0.6061**	**0.7017**	**0.0528**	**10.65**	**(+2.87%)**	**0.4170**	**0.5439**	**0.6754**	**0.0582**	**9.21**	**(+15.17%)**	**0.4728**	**0.6293**	**0.7184**	**0.0542**	**11.19**	**(+1.96%)**

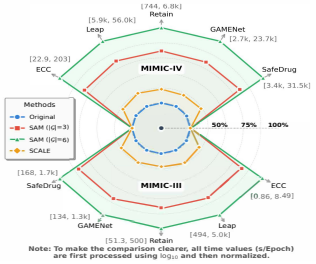


Figure 3: Training time.

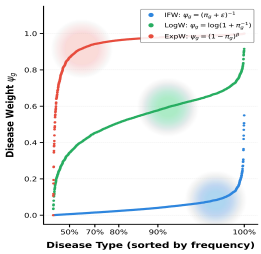
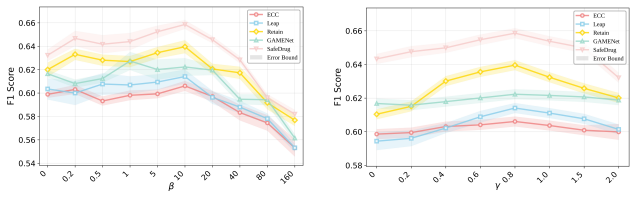


Figure 4: Weight ψ_g.



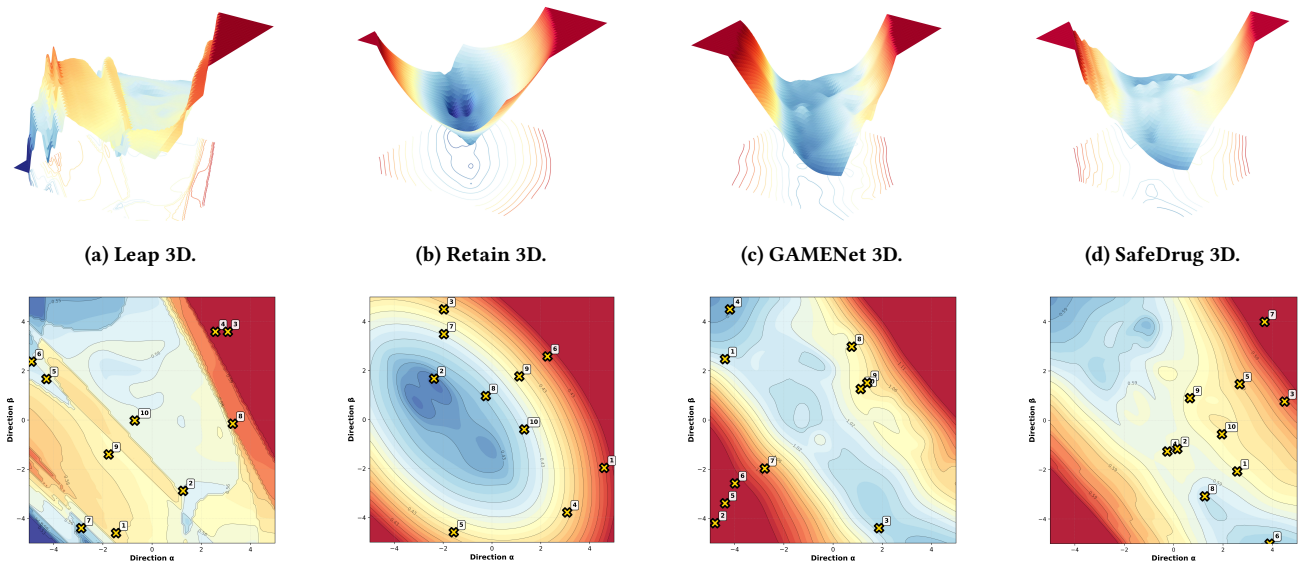
(a) Different β in MIMIC-III. **(b) Different γ in MIMIC-III.**

Figure 5: Ablation study of SCALE w.r.t. β and γ. The shaded area represents the error margin of repeated experiments.

cost is as low as 1.45×. This efficiency stems from the optimized training strategy of SCALE, which avoids the multiple backward passes required by vanilla SAM. In contrast, SAM incurs an approximate time complexity of $|\mathcal{G}|\times$, leading to substantial overhead under large-scale data and high-capacity models. Overall, SCALE achieves a favorable trade-off between performance and efficiency, demonstrating good scalability for real-world CMR applications.

5.2.3 RQ3: Hyperparameter sensitivity (ablation study). We conduct a systematic sensitivity analysis of the three key components of SCALE: the group-wise weight function ψ_g, the sharpness weight γ, and the SAM perturbation magnitude ρ.

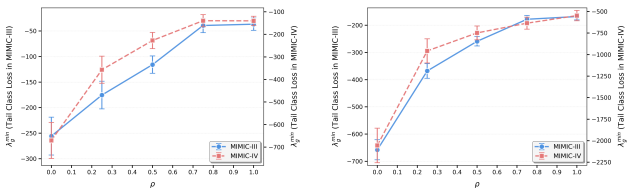
Weight function ψ_g. The weight function ψ_g provides fine-grained group-level control in SCALE. We compare three weighting designs introduced in the main text (Fig. 4). Due to the inherently long-tailed nature of CMR tasks, tail samples are extremely sparse, and overly aggressive weighting may cause overfitting to noisy tail instances, leading to degraded overall performance. Among the three designs, the ExpW function offers a more balanced and stable trade-off. Further sensitivity analysis on the decay rate β (Fig. 5a) shows that moderate increases in β significantly improve recommendation performance by emphasizing tail sharpness and enhancing generalization. However, when β exceeds a threshold (e.g., $\beta > 10$),



(a) Leap 3D. (b) Retain 3D. (c) GAMENet 3D. (d) SafeDrug 3D.

(e) Leap 2D. (f) Retain 2D. (g) GAMENet 2D. (h) SafeDrug 2D.

Figure 6: The loss landscapes of different models for tail classes in the MIMIC-IV dataset are presented in both 2D and 3D views. The color gradient, from red to blue, indicates the loss values from high to low, with yellow X symbols marking the critical points. As shown, compared to before the introduction of SCALE (Fig. 2e and Fig. 2g), SCALE significantly improves the model's loss landscape on tail classes, especially in the saddle point region, where the landscape becomes notably flatter.



(a) Different ρ in SafeDrug. (b) Different ρ in GAMENet.

Figure 7: With high ρ, $\lambda_g^{\min}$ increases, and the model converges to a point with low negative curvature (near minima).

performance deteriorates, suggesting that excessive emphasis on tail sharpness can suppress effective learning for head classes.

Sharpness weight γ. We further analyze the impact of the sharpness weight γ on overall performance. The results in Fig. 5b indicate that moderately increasing γ helps guide the optimization toward flatter minima and improves generalization, leading to an overall performance improvement. In contrast, overly large γ values (e.g., $\gamma > 0.8$) lead to performance degradation, reflecting an inherent trade-off between minimizing empirical loss and suppressing loss landscape sharpness.

SAM perturbation magnitude ρ. The parameter ρ controls the perturbation strength in SAM and plays a critical role in optimization. On MIMIC-III and MIMIC-IV dataset, we analyze the variation of the minimum Hessian eigenvalue $\lambda_g^{\min}$ for three tail classes under different CMR models as ρ increases (Fig. 7). We observe that $\lambda_g^{\min}$ increases with ρ, indicating that stronger perturbations enhance gradient projection along the most negatively curved direction and accelerate escape from saddle points. However, the improvement gradually saturates as ρ becomes large. This behavior arises because the saddle escape rate is jointly determined by the product of ρ and $\lambda_g^{\min}$; when ρ exceeds a certain range (i.e., $\rho > 0.8$), its marginal contribution diminishes, which is consistent with the theoretical results in Theorem 4.1. Overall, SCALE exhibits stable and consistent performance across a wide range of hyperparameter settings, and the empirical observations align with Theorem 4.2.

5.2.4 RQ4: Underlying mechanism of SCALE. To further elucidate the underlying mechanism of SCALE, we conduct a visualization analysis of the loss landscape of SCALE on tail classes using the MIMIC-IV dataset, incorporating both 2D and 3D perspectives and comparing across multiple models, as shown in Fig. 6. The analysis reveals that SCALE exhibits significantly flatter and smoother loss landscapes across all major CMR models, compared to our empirical study in Section 4.1 (as shown in Fig. 2e and Fig. 2g). This phenomenon suggests that SCALE's fine-grained regulatory mechanism effectively alleviates the sharp-minima problem, while its global optimization properties further stabilize the overall optimization process, smoothing out severe fluctuations near saddle points and preventing the model from getting stuck at these points. This empirical result further validates Theorem 4.2.

6 Conclusion

With the widespread use of EHRs, CMR is crucial for personalized healthcare. However, the long-tailed nature of clinical data limits its model generalization to rare diseases, affecting the safety and effectiveness of CMR systems. From an optimization perspective,

severe class imbalance causes gradient descent to stagnate at saddle points with negative curvature, resulting in poor tail-class performance. This issue is worsened by the DDI loss, which adds negative curvature and forces tail classes into non-convex regions. To address these challenges, we propose SCALE, a framework that uses a fine-grained, weighted variant of SAM to balance optimization between head and tail classes. By amplifying gradients in negatively curved directions, SCALE helps escape saddle points and converge to flatter minima, improving rare disease generalization. A tailored weighting strategy prioritizes tail classes without harming head-class performance. Efficient sharpness increment estimation reduces computational overhead. Overall, SCALE improves the reliability and safety of CMR and can be easily integrated into existing CMR frameworks due to its scalability and model-agnostic design.

7 Limitations and Ethical Considerations

This study, despite strong performance on multiple real-world datasets, has several limitations. First, experiments use retrospective EHR data, which may be biased, missing, or inconsistently coded, requiring broader validation in diverse prospective clinical settings. Second, while SCALE improves recommendations for rare diseases, it relies on historical prescribing patterns and cannot replace clinical expertise. Ethically, only de-identified, publicly available medical data are used, ensuring patient privacy.

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A Amplification of Long-Tail Issue by DDI Loss

For simplicity, we omit the sample index and denote $M_t^{(i)}$ as $\boldsymbol{M}$ in the following analysis. To enable second-order analysis, we express the final layer in logit–sigmoid form, $\boldsymbol{M}(\boldsymbol{\theta}) = \boldsymbol{\sigma}(\boldsymbol{z}(\boldsymbol{\theta}))$, where $\boldsymbol{\sigma}(\cdot)$ is the elementwise sigmoid function and $\boldsymbol{z}(\boldsymbol{\theta})$ denotes the pre-activation logits. The DDI loss can then be written as:

$$\mathcal{L}_{\mathrm{DDI}}(\boldsymbol{\theta}; \boldsymbol{M}) = \boldsymbol{\phi}(\boldsymbol{M}(\boldsymbol{\theta})) \triangleq \boldsymbol{M}(\boldsymbol{\theta})^\top \boldsymbol{A} \boldsymbol{M}(\boldsymbol{\theta}), \tag{13}$$

where $\boldsymbol{A}$ is a (possibly approximate) symmetric non-negative matrix encoding DDI structure. Accordingly, the Hessian of the group-conditional loss can be decomposed as

$$\boldsymbol{H}^g_{\mathcal{S}}(\boldsymbol{\theta}) = \boldsymbol{H}^g_{\mathcal{S},\mathrm{Pred}}(\boldsymbol{\theta}) + \alpha \boldsymbol{H}^g_{\mathcal{S},\mathrm{DDI}}(\boldsymbol{\theta}), \tag{14}$$

indicating that whether the DDI loss exacerbates the long-tail problem critically depends on its contribution to the minimum eigenvalue $\lambda_g^{\min}$ of the Hessian.

We analyze the Hessian structure of the single-sample DDI loss. By applying the second-order chain rule for vector-valued functions, the Hessian of $\mathcal{L}_{\mathrm{DDI}}(\boldsymbol{\theta}; \boldsymbol{M})$ can be decomposed as

$$\nabla^2_{\boldsymbol{\theta}} \mathcal{L}_{\mathrm{DDI}}(\boldsymbol{\theta}; \boldsymbol{M}) = \boldsymbol{\mathcal{J}}_{\boldsymbol{M}}^\top \nabla^2_{\boldsymbol{u}} \boldsymbol{\phi}(\boldsymbol{M}(\boldsymbol{\theta})) \boldsymbol{\mathcal{J}}_{\boldsymbol{M}} + \sum_k \frac{\partial \boldsymbol{\phi}}{\partial \boldsymbol{u}_k} \nabla^2_{\boldsymbol{\theta}} M^k, \tag{15}$$

where $\boldsymbol{\mathcal{J}}_{\boldsymbol{M}} = \partial \boldsymbol{M} / \partial \boldsymbol{\theta}$ denotes the Jacobian. Since $\boldsymbol{\phi}(\boldsymbol{u}) = \boldsymbol{u}^\top \boldsymbol{A} \boldsymbol{u}$, we have $\nabla^2_{\boldsymbol{u}} \boldsymbol{\phi} = \boldsymbol{A} + \boldsymbol{A}^\top$. The first term, therefore, forms a Jacobian-weighted quadratic form, which is not guaranteed to be positive definite but is typically not the dominant source of negative curvature. The second term is more critical, as it couples output nonlinearity with network nonlinearity and feeds the DDI structure directly into the curvature of the parameter space. For each component $M^k(\boldsymbol{\theta}) = \sigma(z^k(\boldsymbol{\theta}))$, its second derivative is

$$\nabla^2_{\boldsymbol{\theta}} M^k = \boldsymbol{\sigma}''(z^k) \nabla_{\boldsymbol{\theta}} z^k (\nabla_{\boldsymbol{\theta}} z^k)^\top + \boldsymbol{\sigma}'(z^k) \nabla^2_{\boldsymbol{\theta}} z^k, \tag{16}$$

where $\boldsymbol{\sigma}''(z^k) = M^k(1 - M^k)(1 - 2M^k)$. Substituting into the Hessian expression yields the following decomposition:

$$\boldsymbol{H}_{\mathrm{DDI}} = \underbrace{\sum_k b^k \, \boldsymbol{\sigma}''(z^k) \nabla_{\boldsymbol{\theta}} z^k (\nabla_{\boldsymbol{\theta}} z^k)^\top}_{\text{(I) Sigmoid Saturation Curvature}} + \underbrace{\sum_k b^k \, \boldsymbol{\sigma}'(z^k) \nabla^2_{\boldsymbol{\theta}} z^k}_{\text{(II) Network Intrinsic Curvature}}, \tag{17}$$

where $b^k = \left[(\boldsymbol{A} + \boldsymbol{A}^\top)\boldsymbol{M}\right]^k \geq 0$. This decomposition leads to:

- **Negative curvature from sigmoid saturation.** The outer product $\nabla_{\boldsymbol{\theta}} z^k (\nabla_{\boldsymbol{\theta}} z^k)^\top$ is positive semi-definite, so the sign of the term (I) is entirely determined by $b^k \boldsymbol{\sigma}''(z^k)$. When $M^k > 1/2$, $\boldsymbol{\sigma}''(z^k) < 0$, and since b^k is typically positive under DDI weighting, this term directly induces negative curvature. Thus, sigmoid saturation not only causes gradient vanishing but also introduces systematic negative curvature in the presence of DDI constraints.
- **Amplification of network nonlinearity.** Term (II) arises from $\nabla^2_{\boldsymbol{\theta}} z^k$, which captures the intrinsic second-order nonlinearity of deep networks and is generally indefinite. The DDI-weighted factor $b^k \boldsymbol{\sigma}'(z^k)$ amplifies these indefinite components, further increasing the likelihood of negative curvature directions.

Taken together, this analysis shows that the DDI loss systematically shifts the smallest Hessian eigenvalue $\lambda_g^{\min}$ toward negative values, especially for tail disease groups, making optimization more prone to sharp or negative-curvature regions and thereby exacerbating long-tail generalization degradation.

B Proof of Theorem 4.1

Proof. We apply a first-order Taylor expansion of the gradient:

$$\nabla_{\boldsymbol{\theta}} \hat{\mathcal{L}}^g_{\mathcal{S}}(\boldsymbol{\theta} + \boldsymbol{\varepsilon}) = \nabla_{\boldsymbol{\theta}} \hat{\mathcal{L}}^g_{\mathcal{S}}(\boldsymbol{\theta}) + \boldsymbol{\mathcal{J}} \left(\nabla_{\boldsymbol{\theta}} \hat{\mathcal{L}}^g_{\mathcal{S}}(\boldsymbol{\theta}) \right) \boldsymbol{\varepsilon}. \tag{18}$$

Since the Jacobian of the gradient equals the Hessian,

$$\boldsymbol{\mathcal{J}} \left(\nabla_{\boldsymbol{\theta}} \hat{\mathcal{L}}^g_{\mathcal{S}}(\boldsymbol{\theta}) \right) = \nabla^2_{\boldsymbol{\theta}} \hat{\mathcal{L}}^g_{\mathcal{S}}(\boldsymbol{\theta}) = \boldsymbol{H}^g_{\mathcal{S}}(\boldsymbol{\theta}). \tag{19}$$

Assuming a sufficiently small perturbation radius ρ, the optimal SAM perturbation can be approximated as

$$\boldsymbol{\varepsilon}^\star_g \approx \rho \nabla_{\boldsymbol{\theta}} \hat{\mathcal{L}}^g_{\mathcal{S}}(\boldsymbol{\theta}). \tag{20}$$

Substituting $\boldsymbol{\varepsilon}_g^\star$ into Eq. (18), the SAM gradient admits the approximation

$$\nabla_{\boldsymbol{\theta}}\hat{\mathcal{L}}_{\mathcal{S}}^{g,\mathrm{SAM}}(\boldsymbol{\theta}) = \nabla_{\boldsymbol{\theta}}\hat{\mathcal{L}}_{\mathcal{S}}^{g}(\boldsymbol{\theta}+\boldsymbol{\varepsilon}_g^\star) \approx \nabla_{\boldsymbol{\theta}}\hat{\mathcal{L}}_{\mathcal{S}}^{g}(\boldsymbol{\theta}) + \rho \boldsymbol{H}_{\mathcal{S}}^{g}(\boldsymbol{\theta})\nabla_{\boldsymbol{\theta}}\hat{\mathcal{L}}_{\mathcal{S}}^{g}(\boldsymbol{\theta}). \tag{21}$$

We next analyze the second moment of the SAM gradient projected onto the direction of most negative curvature $\boldsymbol{v}_g(\boldsymbol{\theta})$. Using the eigenvector property $\boldsymbol{v}_g(\boldsymbol{\theta})^\top \boldsymbol{H}_{\mathcal{S}}^{g}(\boldsymbol{\theta}) = \lambda_g^{\min}\boldsymbol{v}_g(\boldsymbol{\theta})^\top$, the desired result follows.

$$\begin{aligned}
&\mathbb{E}\left[\langle \boldsymbol{v}_g(\boldsymbol{\theta}), \nabla_{\boldsymbol{\theta}}\hat{\mathcal{L}}_{\mathcal{S}}^{g,\mathrm{SAM}}(\boldsymbol{\theta})\rangle^2\right] \\
&= \mathbb{E}\left[\left(\langle \boldsymbol{v}_g(\boldsymbol{\theta}), \nabla_{\boldsymbol{\theta}}\hat{\mathcal{L}}_{\mathcal{S}}^{g}(\boldsymbol{\theta}) + \rho \boldsymbol{H}_{\mathcal{S}}^{g}(\boldsymbol{\theta})\nabla_{\boldsymbol{\theta}}\hat{\mathcal{L}}_{\mathcal{S}}^{g}(\boldsymbol{\theta})\rangle\right)^2\right] \\
&= \mathbb{E}\left[\left(\langle \boldsymbol{v}_g(\boldsymbol{\theta}), \nabla_{\boldsymbol{\theta}}\hat{\mathcal{L}}_{\mathcal{S}}^{g}(\boldsymbol{\theta})\rangle + \rho\langle \boldsymbol{v}_g(\boldsymbol{\theta}), \boldsymbol{H}_{\mathcal{S}}^{g}(\boldsymbol{\theta})\nabla_{\boldsymbol{\theta}}\hat{\mathcal{L}}_{\mathcal{S}}^{g}(\boldsymbol{\theta})\rangle\right)^2\right] \\
&= \mathbb{E}\left[\left(\langle \boldsymbol{v}_g(\boldsymbol{\theta}), \nabla_{\boldsymbol{\theta}}\hat{\mathcal{L}}_{\mathcal{S}}^{g}(\boldsymbol{\theta})\rangle + \rho\boldsymbol{v}_g(\boldsymbol{\theta})^\top \boldsymbol{H}_{\mathcal{S}}^{g}(\boldsymbol{\theta})\nabla_{\boldsymbol{\theta}}\hat{\mathcal{L}}_{\mathcal{S}}^{g}(\boldsymbol{\theta})\right)^2\right] \\
&= \mathbb{E}\left[\left(\langle \boldsymbol{v}_g(\boldsymbol{\theta}), \nabla_{\boldsymbol{\theta}}\hat{\mathcal{L}}_{\mathcal{S}}^{g}(\boldsymbol{\theta})\rangle + \rho\lambda_g^{\min}\langle \boldsymbol{v}_g(\boldsymbol{\theta}), \nabla_{\boldsymbol{\theta}}\hat{\mathcal{L}}_{\mathcal{S}}^{g}(\boldsymbol{\theta})\rangle\right)^2\right] \\
&= \mathbb{E}\left[\left((1+\rho\lambda_g^{\min})\langle \boldsymbol{v}_g(\boldsymbol{\theta}), \nabla_{\boldsymbol{\theta}}\hat{\mathcal{L}}_{\mathcal{S}}^{g}(\boldsymbol{\theta})\rangle\right)^2\right],
\end{aligned} \tag{22}$$

where $\langle \boldsymbol{a}, \boldsymbol{b}\rangle = \boldsymbol{a}^\top\boldsymbol{b}$ is the dot product. Factoring out the constant term yields

$$\mathbb{E}\left[\langle \boldsymbol{v}_g(\boldsymbol{\theta}), \nabla_{\boldsymbol{\theta}}\hat{\mathcal{L}}_{\mathcal{S}}^{g,\mathrm{SAM}}(\boldsymbol{\theta})\rangle^2\right] = (1+\rho\lambda_g^{\min})^2\mathbb{E}\left[\langle \boldsymbol{v}_g(\boldsymbol{\theta}), \nabla_{\boldsymbol{\theta}}\hat{\mathcal{L}}_{\mathcal{S}}^{g}(\boldsymbol{\theta})\rangle^2\right]. \tag{23}$$

which is exactly the statement of Theorem 4.1. □

C Proof of Theorem 4.2

Before proving Theorem 4.2, we introduce key preliminaries that relate the expected loss to its empirical counterpart. The following lemma establishes this connection by characterizing the role of curvature in controlling generalization.

C.1 Preliminary Proof

Lemma C.1 ([The reformulation of Lemma in [46]). *] Fix any $\rho > 0$ and define $\varphi = \rho/(\sqrt{d}+\sqrt{2\ln n})$. There exists a remainder term $\tilde{O}\left(\frac{d\rho^2}{(\sqrt{d}+\sqrt{2\ln n})^2}\right)$ such that, with probability at least $1-\delta$ over $\mathcal{S}\sim\mathcal{D}$,*

$$\mathcal{L}_{\mathcal{D}}^{\gamma}(\boldsymbol{\theta}) \le \max_{\|\boldsymbol{\varepsilon}\|_2\le\rho} 2\hat{\mathcal{L}}_{\mathcal{S}}^{\gamma}(\boldsymbol{\theta}+\boldsymbol{\varepsilon}) - \frac{\rho^2}{2\left(\sqrt{d}+\sqrt{2\ln n}\right)^2}\mathrm{Tr}\big(\boldsymbol{H}_{\mathcal{D}}(\boldsymbol{\theta})\big) + \frac{C(\boldsymbol{\theta},\rho,d,n,\delta)}{n} - \tilde{O}\Big(\tfrac{d\rho^2}{(\sqrt{d}+\sqrt{2\ln n})^2}\Big). \tag{24}$$

where the complexity term $C(\boldsymbol{\theta},\rho,d,n,\delta)$ is the same as in Lemma C.3.

Lemma C.2 (The reformulation of Theorem in [3]). *With probability at least $1-\delta$ over the draw of the training set $\mathcal{S}\sim\mathcal{D}$,*

$$\Phi_{\mathcal{D}}^{\gamma}(\boldsymbol{\theta}) \le 2\Phi_{\mathcal{S}}^{\gamma}(\boldsymbol{\theta}) + \frac{40(\xi+\gamma\hat{\xi})}{3n}\log\frac{2}{\delta}, \tag{25}$$

where $\Phi_{\mathcal{D}}^{\gamma}(\boldsymbol{\theta}) = \mathcal{L}_{\mathcal{D}}(\boldsymbol{\theta}) - \gamma\mathcal{L}_{\mathcal{D}}^{\gamma}(\boldsymbol{\theta})$ and $\Phi_{\mathcal{S}}^{\gamma}(\boldsymbol{\theta}) = \hat{\mathcal{L}}_{\mathcal{S}}(\boldsymbol{\theta}) - \gamma\hat{\mathcal{L}}_{\mathcal{S}}^{\gamma}(\boldsymbol{\theta})$.

Lemma C.3 (The reformulation of Theorem in [12]). *Fix $\rho > 0$ and define $\varphi = \rho/(\sqrt{d}+\sqrt{2\ln n})$. With probability at least $1-\delta$ over the draw of $\mathcal{S}\sim\mathcal{D}$,*

$$\mathbb{E}_{\boldsymbol{\varepsilon}\sim\mathcal{N}(0,\varphi^2\mathbf{I})}\left[\mathcal{L}_{\mathcal{D}}^{\gamma}(\boldsymbol{\theta}+\boldsymbol{\varepsilon})\right] \le \max_{\|\boldsymbol{\varepsilon}\|_2\le\rho} 2\hat{\mathcal{L}}_{\mathcal{S}}^{\gamma}(\boldsymbol{\theta}+\boldsymbol{\varepsilon}) + \frac{C(\boldsymbol{\theta},\rho,d,n,\delta)}{n}, \tag{26}$$

where the complexity term is given by

$$\begin{aligned}
C(\boldsymbol{\theta},\rho,d,n,\delta) =& 2 + 2\hat{\xi} + 2d\ln\Big(1+\frac{\|\boldsymbol{\theta}\|_2^2}{d\rho^2}\Big) + 4d\log\Big(\sqrt{d}+\sqrt{2\ln n}\Big) \\
&+ 4\log\left(\frac{\pi^2\sqrt{n}\big(1+n\hat{\xi}\big)^2}{3\delta}\right).
\end{aligned} \tag{27}$$

C.2 Formal Proof

Proof. By definition, for any $\gamma \ge 0$,

$$\mathcal{L}_{\mathcal{D}}(\boldsymbol{\theta}) = \Phi_{\mathcal{D}}^{\gamma}(\boldsymbol{\theta}) + \gamma\mathcal{L}_{\mathcal{D}}^{\gamma}(\boldsymbol{\theta}). \tag{28}$$

Applying Lemma C.2 to $\Phi_{\mathcal{D}}^{\gamma}(\boldsymbol{\theta})$ and Lemma C.1 to $\mathcal{L}_{\mathcal{D}}^{\gamma}(\boldsymbol{\theta})$, we obtain

$$\begin{aligned}
\mathcal{L}_{\mathcal{D}}(\boldsymbol{\theta}) \le\ & 2\Phi_{\mathcal{S}}^{\gamma}(\boldsymbol{\theta}) + \frac{40(\xi+\gamma\hat{\xi})}{3n}\ln\frac{2}{\delta} \\
&+ \gamma\left[\max_{\|\boldsymbol{\varepsilon}\|_2\le\rho} 2\hat{\mathcal{L}}_{\mathcal{S}}^{\gamma}(\boldsymbol{\theta}+\boldsymbol{\varepsilon}) - \frac{\rho^2}{2(\sqrt{d}+\sqrt{2\ln n})^2}\mathrm{Tr}\big(\boldsymbol{H}_{\mathcal{D}}(\boldsymbol{\theta})\big) + \frac{C(\boldsymbol{\theta},\rho,d,n,\delta)}{n} - \tilde{O}\Big(\tfrac{d\rho^2}{(\sqrt{d}+\sqrt{2\ln n})^2}\Big)\right].
\end{aligned} \tag{29}$$

Using $\Phi_{\mathcal{S}}^{\gamma}(\boldsymbol{\theta}) = \hat{\mathcal{L}}_{\mathcal{S}}(\boldsymbol{\theta}) - \gamma\hat{\mathcal{L}}_{\mathcal{S}}^{\gamma}(\boldsymbol{\theta})$, the empirical terms can be regrouped as

$$\begin{aligned}
&2\Phi_{\mathcal{S}}^{\gamma}(\boldsymbol{\theta}) + \gamma\max_{\|\boldsymbol{\varepsilon}\|_2\le\rho} 2\hat{\mathcal{L}}_{\mathcal{S}}^{\gamma}(\boldsymbol{\theta}+\boldsymbol{\varepsilon}) \\
&= \frac{2}{|\mathcal{G}|\pi_{\min}}\Big[|\mathcal{G}|\pi_{\min}\hat{\mathcal{L}}_{\mathcal{S}}(\boldsymbol{\theta}) + \gamma|\mathcal{G}|\pi_{\min}\hat{\mathcal{L}}_{\mathcal{S}}^{\ln}(\boldsymbol{\theta})\Big] \\
&= \frac{2}{|\mathcal{G}|\pi_{\min}}\mathcal{T}_{\mathcal{S}}(\boldsymbol{\theta}).
\end{aligned} \tag{30}$$

where $\mathcal{T}_{\mathcal{S}}(\boldsymbol{\theta}) = \hat{\mathcal{L}}_{\mathcal{S}}(\boldsymbol{\theta}) + \gamma\hat{\mathcal{L}}_{\mathcal{S}}^{\ln}(\boldsymbol{\theta})$ is exactly the SCALE objective. Finally, since the imbalanced training distribution lower-bounds the balanced one, $\mathcal{L}_{\mathcal{D}}(\boldsymbol{\theta}) \ge |\mathcal{G}|\pi_{\min}\mathcal{L}_{\mathcal{D}_{\mathrm{bal}}}(\boldsymbol{\theta})$, we obtain

$$\mathcal{L}_{\mathcal{D}_{\mathrm{bal}}}(\boldsymbol{\theta}) \le \frac{1}{|\mathcal{G}|\pi_{\min}}\mathcal{L}_{\mathcal{D}}(\boldsymbol{\theta}). \tag{31}$$

Dividing the previous upper bound on $\mathcal{L}_{\mathcal{D}}(\boldsymbol{\theta})$ by $|\mathcal{G}|\pi_{\min}$ yields, with probability at least $1-\delta$,

$$\begin{aligned}
\mathcal{L}_{\mathcal{D}_{\mathrm{bal}}}(\boldsymbol{\theta}) \le\ & \frac{2}{|\mathcal{G}|\pi_{\min}}\mathcal{T}_{\mathcal{S}}(\boldsymbol{\theta}) - \frac{\gamma\rho^2}{2|\mathcal{G}|\pi_{\min}(\sqrt{d}+\sqrt{2\ln n})^2}\mathrm{Tr}\big(\boldsymbol{H}_{\mathcal{D}}(\boldsymbol{\theta})\big) \\
&+ \frac{1}{|\mathcal{G}|\pi_{\min}}\left[\frac{40(\xi+\gamma\hat{\xi})}{3n}\ln\frac{2}{\delta} + \frac{\gamma C(\boldsymbol{\theta},\rho,d,n,\delta)}{n}\right] \\
&- \frac{\gamma}{|\mathcal{G}|\pi_{\min}}\tilde{O}\Big(\tfrac{d\rho^2}{(\sqrt{d}+\sqrt{2\ln n})^2}\Big),
\end{aligned} \tag{32}$$

which completes the proof of Theorem 4.2. □